Project Report: Domestic Wiring Demonstration Board for a Village Skill Centre

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Aim

To build a wall-mounted demonstration board showing a lighting circuit, a fan circuit with regulator and a 6 A / 16 A socket circuit, each protected by its own MCB, together with a wiring diagram and a safety checklist, so that village youth at the skill centre can learn safe domestic wiring.

Materials and Tools

18 mm plywood board 4 ft x 3 ft, PVC casing-capping 20 mm (8 m), 4-way SPN distribution board.

MCBs: 1 x 32 A DP main, 2 x 6 A SP (light, fan), 1 x 16 A SP (power socket); 30 mA RCCB 25 A.

1.5 sq mm and 2.5 sq mm FR copper wire (red, black, green), 2 x 6 A switches, 1 x 16 A switch, 1 x 6 A socket, 1 x 16 A socket, fan regulator, LED batten, 2-way switch pair.

Tools: combination plier, wire stripper, screwdriver set, drill machine, multimeter, 500 V insulation tester.

Procedure

1. Drew the layout and single-line diagram on paper and got it checked by the instructor.
2. Marked and fixed the casing-capping, switchboards and DB on the plywood with screws every 30 cm.
3. Ran 1.5 sq mm wire for light and fan circuits and 2.5 sq mm for the 16 A socket circuit.
4. Wired the 2-way staircase light circuit, the fan through the regulator, and both sockets with earth.
5. Connected each circuit to its own MCB in the DB, through the RCCB and the main MCB.
6. Did a continuity test of every circuit and a polarity test so that switches break the phase wire.
7. Did an insulation resistance test at 500 V between phase-neutral, phase-earth and neutral-earth.
8. Powered the board from a 3-pin top with a 16 A plug only after the instructor checked it.

Safety Precautions

Board never connected to supply during wiring; supply plug kept with the instructor until final test.

Correct colour code used: red phase, black neutral, green earth; all metal parts earthed.

RCCB tested with its test button; MCB ratings chosen for wire size (6 A on 1.5 sq mm, 16 A on 2.5 sq mm).

No bare conductor outside terminals; all screws tightened; insulated tools and rubber mat used.

A printed safety checklist (8 points) is fixed on the board for learners.

Observations and Results

Insulation resistance: phase-earth 120 megohm, neutral-earth 95 megohm, phase-neutral 150 megohm (minimum 1 megohm).

Supply voltage 231 V. LED batten current 0.09 A; fan at full speed 0.31 A; 16 A socket with 1 kW heater 4.3 A.

RCCB tripped in under 0.1 s when the test button was pressed (3 trials).

All 5 circuits worked first time after testing; the 2-way switch works from both ends.

Conclusion

The demonstration board is complete, tested and safe. It shows lighting, fan and socket circuits with

MCB and RCCB protection exactly as in a real house. Doing the insulation and polarity tests showed me why every installation must be tested before it is energised. The diagram and checklist will help the centre teach.

Logbook and Bill of Materials

Logbook: 22 hours over 4 days (planning and diagram 4 h, fixing 6 h, wiring 8 h, testing and handover 4 h).

BOM: plywood Rs. 1,150, casing-capping Rs. 240, DB Rs. 450, MCBs and RCCB Rs. 1,900, wires Rs. 780, switches, sockets, regulator and batten Rs. 1,050. Total Rs. 5,570.

The wiring diagram (A3) and the safety checklist were handed over with the board.